SEMESTER - I

COURSE: MAJOR – 1(MJ-1) **TOTAL CREDITS:** THEORY-03, PRACTICAL-01
PAPER NAME: MECHANICS AND PROPERTIES OF MATTER **TEACHING HOURS:** THEORY-45, PRACTICAL-30

EVALUATION			
	External Exam	**Internal Exam**	**Practical**
Full Marks	**60**	15 (10 Written + 5 Attendance/Overall Class Performance)	**25**
Duration of Exam	3 Hours	1 Hour	6 Hours
Pass Marks	30 Marks		10 marks

COURSE OBJECTIVES:

This course aims to enable the students to acquire the key concepts of the general properties of matter, the motion of a particle under central force field, oscillations and special theory of relativity.

COURSE OUTCOMES:

1. Learn about the behavior of physical bodies around us in daily life.
2. Understand the dynamics of planetary motion.
3. Build a foundation of various applied field in science and technology.
4. Develop the analytical thinking on Mechanics in order to understand the response of the classical systems to external forces.

PART 'A'

THEORY COURSE CONTENTS:

UNIT-I:

General Properties of Matter (15 Lectures): Hooke's law. Stress-strain diagram, Elastic moduli, Poisson's Ratio: Expression for Poisson's ratio in terms of elastic constants. Relation between Elastic constants. Kinematics of Moving Fluids: Viscous fluid, Poiseuille's Equation for Flow of a Liquid through a Capillary Tube with correction, Flow of compressible fluid through a capillary tube, Rankine's methods for measurement of viscosity of gas. Surface tension and surface energy, angle of contact, Expression for excess pressure, Principle of virtual work, Ripples and Gravity waves.

UNIT-II:

Central Force Motion and Oscillation (15 Lectures): Motion of a particle under a central force field. Conservation of angular momentum. Kepler's Laws of planetary motion and their derivations. Satellite in circular orbit and applications, Centrifugal force. Simple Harmonic Oscillations (SHM).Differential equation of SHM and its solution. Kinetic energy, potential energy, total energy and their time-average values. Examples of Physical Systems Executing SHM: Simple Pendulum, Compound Pendulum, Torsional Pendulum, LC-Circuit. Damped oscillation. Forced oscillations: Resonance, sharpness of resonance, power dissipation and Quality Factor.

UNIT-III:

Special Theory of Relativity (15 Lectures): Inertial and Non-inertial frames. Michelson-Morley Experiment and its outcome. Postulates of Special Theory of Relativity. Lorentz Transformations. Lorentz contraction. Time dilation. Simultaneity and order of events. Relativistic addition of velocities. Variation of mass with velocity. Massless Particles. Mass-energy Equivalence. Relativistic Doppler Effect.

SUGGESTED READINGS:

1. An introduction to mechanics, D. Kleppner, R.J. Kolenkow, 1973, McGraw-Hill.
2. Mechanics by D. S. Mathur, S. Chand.
3. Physics, Resnick, Halliday and Walker, 2008, Wiley.
4. Feynman Lectures, Vol. I, R.P. Feynman, R.B. Leighton, M. Sands, 2008, Pearson Education
5. Introduction to Special Relativity, R. Resnick, 2005, John Wiley and Sons.
6. University Physics, Ronald Lane Reese, 2003, Thomson Brooks/Cole.

PART 'B'

PRACTICAL COURSE CONTENTS:

List of Practical:

1. To determine the value of g using simple Pendulum
2. To determine the value of g using Bar Pendulum.
3. To determine the value of g using Kater's Pendulum.
4. To study the Motion of Spring and calculate (a) Spring constant, (b) g and (c) Modulus of rigidity.
5. To determine the elastic Constants of a wire by Searle's method.
6. To determine the Modulus of Rigidity of a Wire by Maxwell's needle.

SUGGESTED READINGS:

1. Advanced Practical Physics for students, B.L. Flint and H.T. Worsnop, 1971, Asia Publishing House
2. A Text Book of Practical Physics, I. Prakash& Ramakrishna, 11th Ed., 2011, Kitab Mahal
3. Advanced level Physics Practicals, Michael Nelson and Jon M. Ogborn, 4th Edition, reprinted 1985, Heinemann Educational Publishers.
4. A Laboratory Manual of Physics for undergraduate classes, D.P. Khandelwal, 1985, Vani Pub.

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